Recursion

Repeat Me Smaller!

CS 101: Programming for Future Advanced Engineers

Mastering Recursion in 45 Minutes

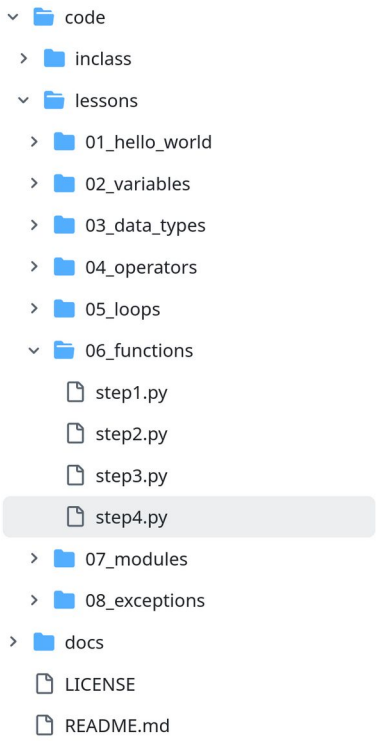
- Recap
- What Is Recursion?
- Why Use Recursion?
- Think Recursively
- When to **Avoid** Recursion
- When to **Use** Recursion

Big problem → Smaller problem → Even smaller problem

Recap

https://github.com/amirih/cs-101-2026-spring

- We are in **step4.py** of the **06_functions** folder




```python
# recursion examples
# recursion is a programming technique where a function calls itself in order to
# solve a problem. It typically involves a base case that stops the recursion and
# a recursive case that breaks the problem into smaller subproblems.

def sum_to_n(n):
    # Base case: if n is 1, return 1
    if n == 1:
        return 1
    else:
        # Recursive case: return n plus the sum of the first n-1 natural numbers
        return n + sum_to_n(n - 1)

def factorial(n):
    # Base case: if n is 0, return 1
    if n == 0:
        return 1
    else:
        # Recursive case: return n multiplied by the factorial of n-1
        return n * factorial(n - 1)

def fibonacci(n):
    # Base case: if n is 0, return 0
    if n == 0:
        return 0
    # Base case: if n is 1, return 1
    elif n == 1:
        return 1
    else:
        # Recursive case: return the sum of the two preceding Fibonacci numbers
        return fibonacci(n - 1) + fibonacci(n - 2)
```

Recap

- A **function** or **method** is a reusable block of code that performs a specific task
- It can receive inputs called **parameters** or **arguments**
- A function may **return** a value, or it may simply perform an action

```python
def print_hello():
    print("Hello, world!")

if __name__ == '__main__':

    print_hello()
```

Recap

```python
def with_parameters_no_return(a, b, c):
    print(f"a: {a}, b: {b}, c: {c}")

def with_parameters_with_return(a, b, c):
    print(f"a: {a}, b: {b}, c: {c}")
    return a + b + c

def no_parameters_no_return():
    print("This function has no parameters")
    print("and it does not return anything.")

def no_parameters_with_return():
    print("This function has no parameters")
    print("but it returns a message.")
    return "Hello, World!"
```

```python
if __name__ == '__main__':

    with_parameters_no_return(1, 2, 3)

    result = with_parameters_with_return(1, 2, 3)
    print(f"Result: {result}")

    no_parameters_no_return()

    message = no_parameters_with_return()
    print(f"Message: {message}")
```

Recap

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Recap

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if __name__ == '__main__':

    with_parameters_no_return(1, 2, 3)

    result = with_parameters_with_return(1, 2, 3)
    print(f"Result: {result}")

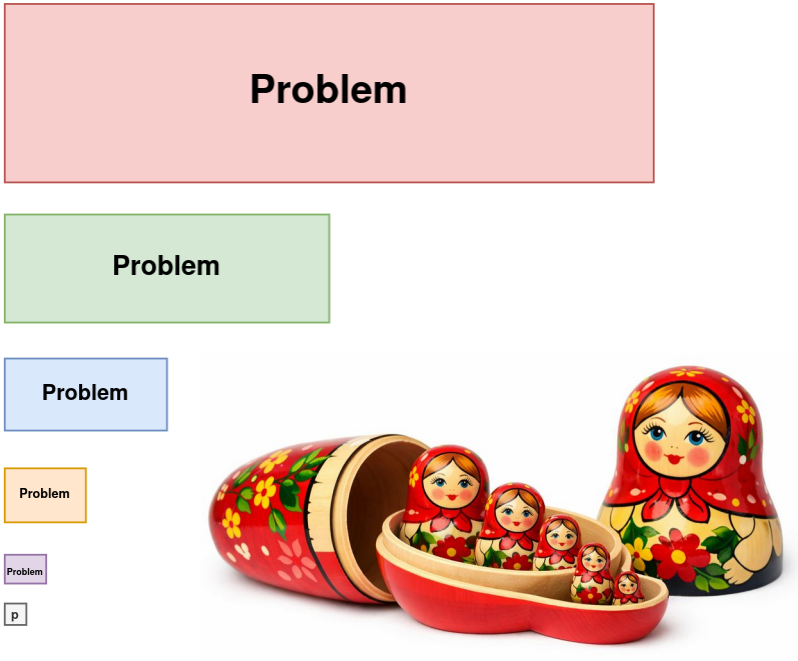
    no_parameters_no_return()

    message = no_parameters_with_return()
    print(f"Message: {message}")
```

What Is Recursion?

- Recursion is a way of solving a problem by **solving a smaller version** of the same problem
- Think of a **Russian doll** as a *mental model* for recursion.

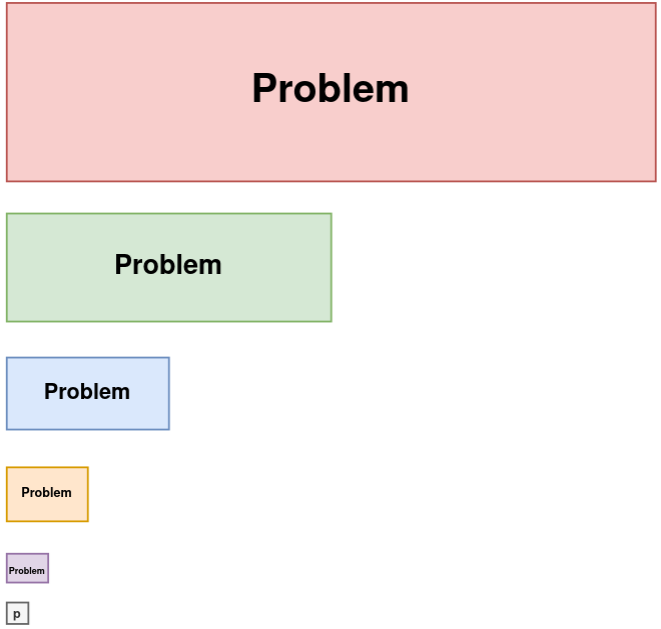
  Inside one doll is a smaller doll, and inside that is an even smaller one, until you reach the smallest doll



Recursion: *Type 1*

- Solve the problem by reducing it to a **single smaller problem**
- Each step (recursive call) makes the **problem smaller**
- The process stops at the smallest possible case, the **base case**

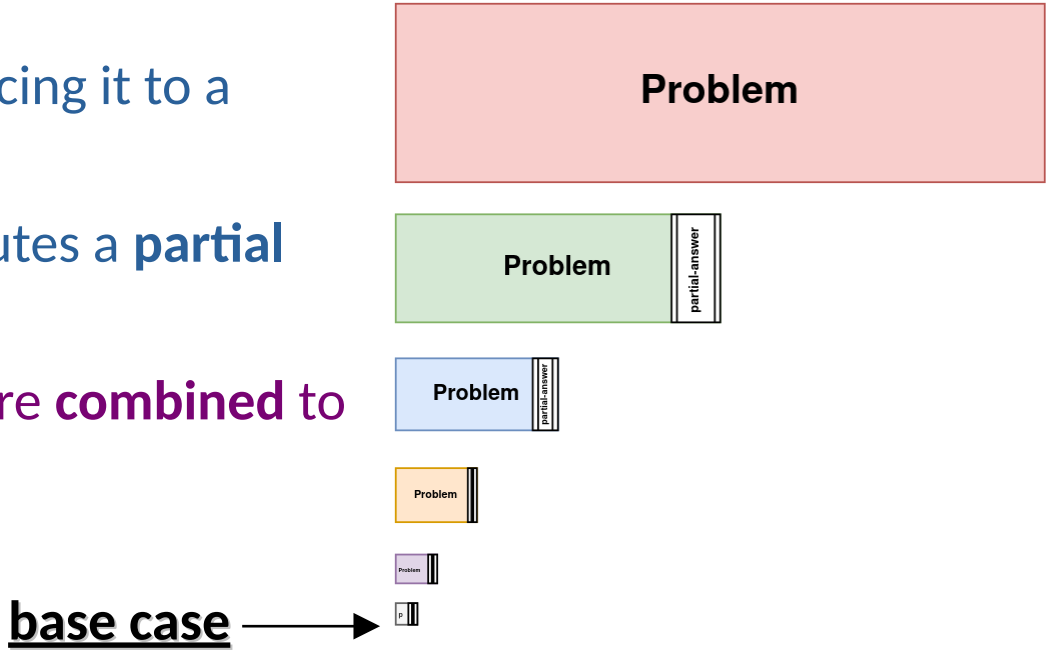
One recursive call
No partial answer



Recursion: *Type 2*

- Solve the problem by reducing it to a **single smaller problem**
- Each recursive call contributes a **partial answer**
- Then the partial answers are **combined** to produce the final solution

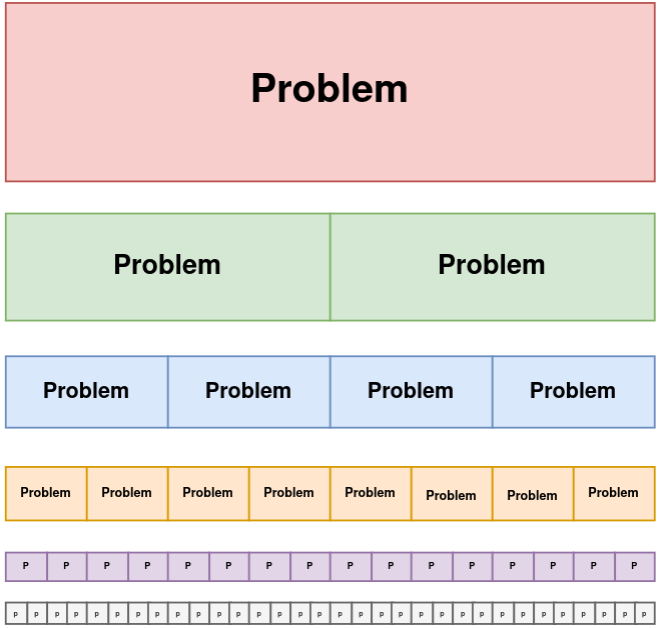
One recursive call
With a partial answer



Recursion: *Type 3*

- Solve the problem by breaking it into **<u>multiple</u> smaller problems**

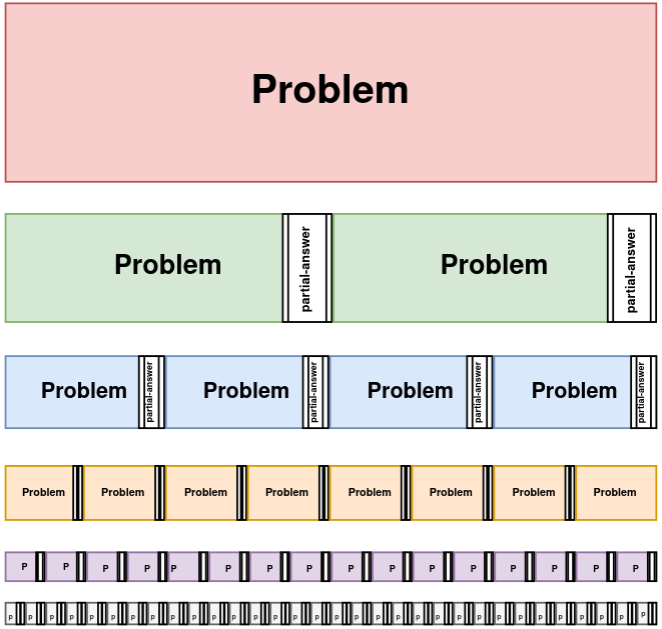
<u>Multiple</u> recursive calls,
<u>No</u> partial answer



Recursion: *Type 4*

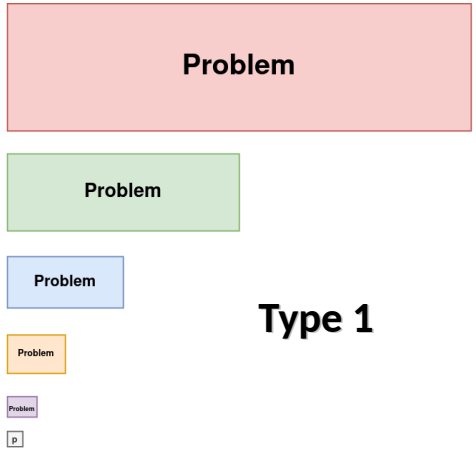
- Solve the problem by breaking it into **multiple smaller problems**
- Each recursive call contributes a **partial answer**

Multiple recursive calls, With partial answers

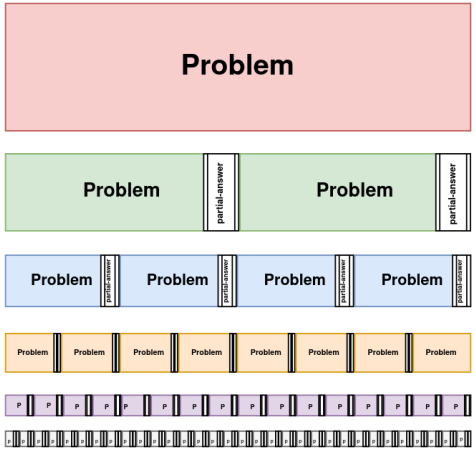


Comparing Types of Recursion

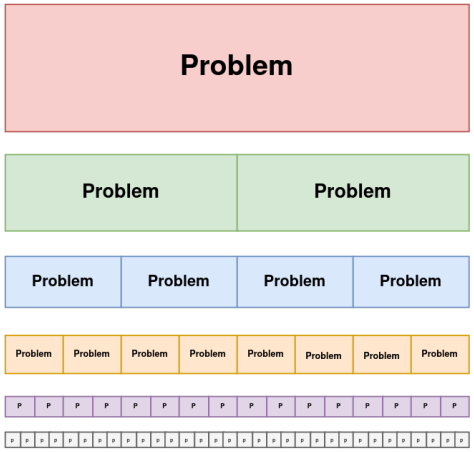
- Type 1: <u>One</u> recursive call — <u>No</u> partial answer
- Type 2: <u>One</u> recursive call — <u>With</u> a partial answer
- Type 3: <u>Multiple</u> recursive calls — <u>No</u> partial answer
- Type 4: <u>Multiple</u> recursive calls — <u>With</u> partial answers



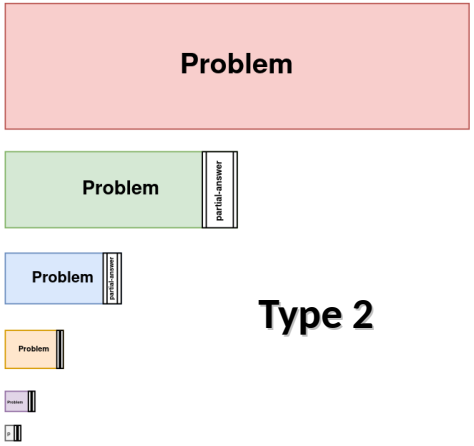
Type 1



Type 4



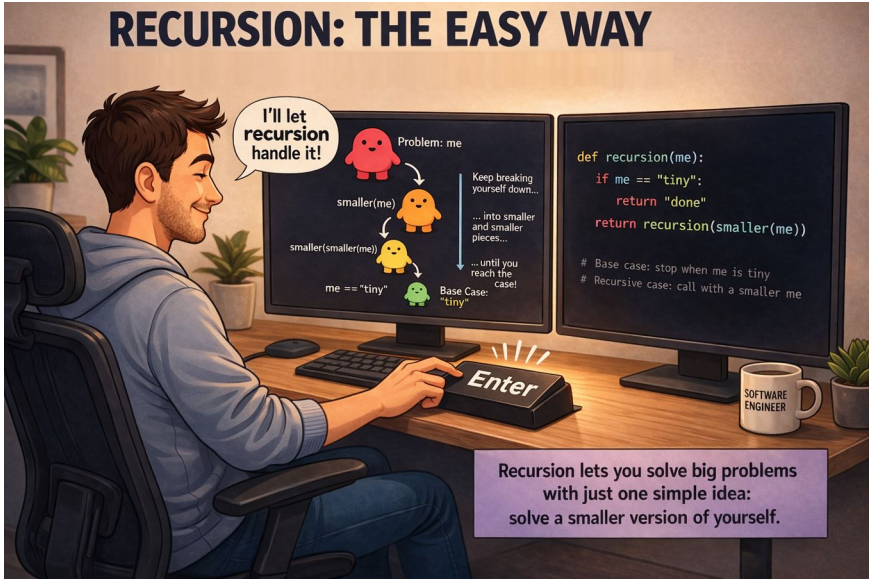
Type 3



Type 2

Why Do We Need Recursion?

- **Some** problems are **easier** to solve with recursion
- Recursion helps us write solutions that are **organized** and **easier** to understand



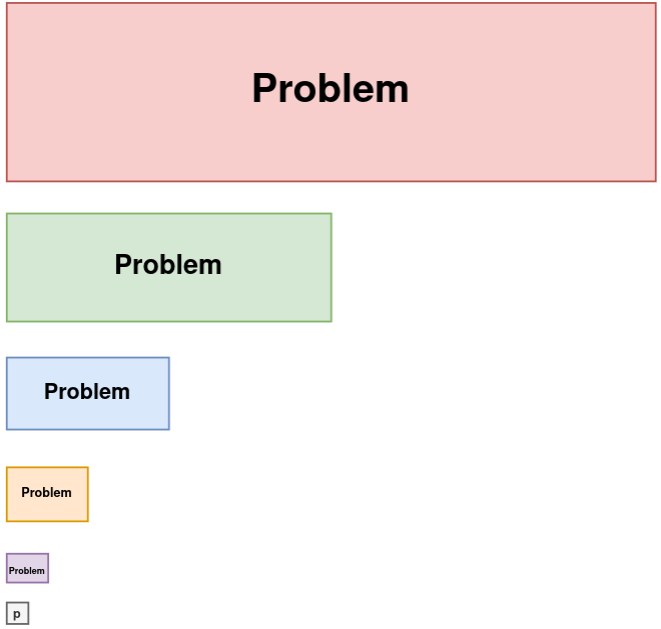
Think Recursively: *Key Idea of Type 1*

- **One recursive** call at each step
- **No partial answer** is kept
- The goal is usually to **finish a process**

```
function solve(problem):
    # base case
    if problem is smallest:
        return answer

    # recursive case
    do_one_step()

    return solve(smaller_problem)
```



One recursive call
No partial answer

Think Recursively: *Count Down*

- Print the current number
- Count down the **remaining numbers**
- **Stop** when you reach zero

10, 9, 8, 7, … , 1

```
function countdown(n):
    # base case
    if n == 0:
        return

    # recursive case
    print(n)

    return countdown(n - 1)
```

Think Recursively: *Climb Stairs*

- Take one step
- Climb the remaining stairs
- **Stop** when no stairs are left

```
function climb(stairs):
    # base case
    if stairs == 0:
        return

    # recursive case
    take_one_step()

    return climb(stairs - 1)
```

Think Recursively: *Empty a Basket*

- Remove one item
- Empty the rest
- **Stop** when the basket is empty

```
function empty_basket(items):
    # base case
    if items == 0:
        return

    # recursive case
    remove_one_item()

    return empty_basket(items - 1)
```

Challenge Time!

Challenge 1: *Count down*

Problem: Starting from a number and going backward by 1 until you reach 0.

Your goal: Implement `countdown(n)`, a recursive function that counts down to 1.

Key idea:

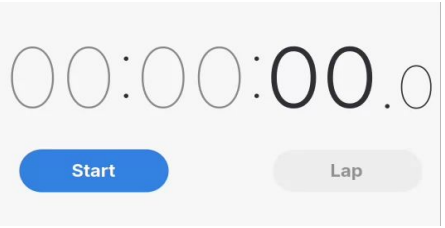
- **Base case:** when `n == 0`, stop and return
- **Recursive case:** print `n`, then call `countdown(n - 1)`

```
print(---)
return countdown(------)
```

Examples `countdown(2)`:

```
2
1
```

Challenge Time!



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Key idea:

- **Base case:** when `n == 0`, stop and return
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```
print(---)
return countdown(------)
```

Examples `countdown(2)`: 2

1

Challenge Time!

Challenge 1 Solved!

```python
def countdown(n):

    if n == 0:
        return

    print(n)

    return countdown(n - 1)

if __name__ == "__main__":

    countdown(5)
```

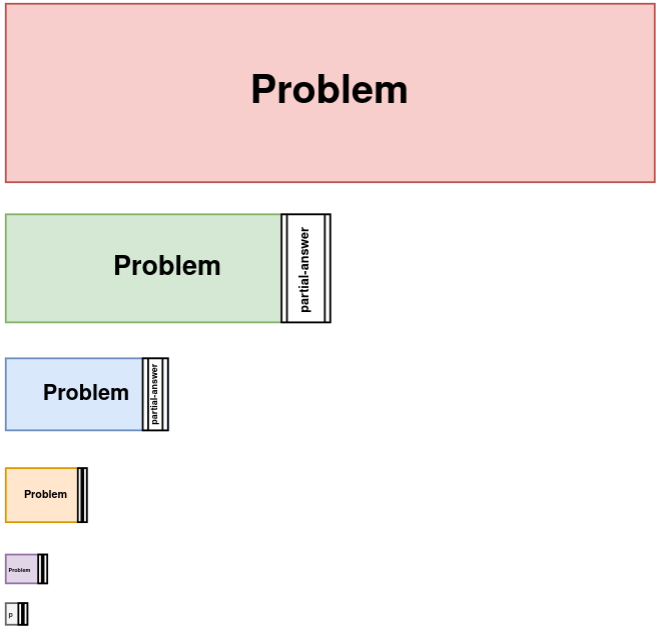
```
Output:
5
4
3
2
1
```

Think Recursively: *Key Idea of Type 2*

- **One** recursive call at each step
- Each level contributes **part of the solution**
- The final answer is built by **combining** these partial results

```
function solve(problem):
    # base case
    if problem is smallest:
        return base_answer

    # recursive case
    return partial_answer + solve(smaller_problem)
```



One recursive call
With a partial answer

Think Recursively: *Sum Numbers from 1 to n*

- Keep the current number
- Add it to the sum of the smaller numbers
- **Stop** at 1 and return 1

```
sum(4) = 4 + sum(3)
sum(3) = 3 + sum(2)
sum(2) = 2 + sum(1)
sum(1) = 1

sum(4) = 4 + (3 + (2 + 1))
```

```
function sum(n):
    # base case
    if n == 1:
        return 1

    # recursive case
    return n + sum(n - 1)
```

Think Recursively: *Factorial*

- Keep the current number
- Multiply it by the factorial of the smaller number
- **Stop** at 1 and return 1

```
factorial(4) = 4 * factorial(3)
factorial(3) = 3 * factorial(2)
factorial(2) = 2 * factorial(1)
factorial(1) = 1

factorial(4) = 4 * (3 * (2 * 1))
```

```
function factorial(n):
    # base case
    if n == 1:
        return 1

    # recursive case
    return n * factorial(n - 1)
```

Think Recursively: *String of Stars*

- Keep one star
- Add it to the smaller result
- **Stop** at zero and return “”

```
function stars(n):
    # base case
    if n == 0:
        return ""

    # recursive case
    return "*" + stars(n - 1)
```

```
stars(3) = '*' + stars(2)
stars(2) = '*' + stars(1)
stars(1) = '*' + stars(0)
stars(0) = ''
stars(3) = '*' + ('*' + ('*' + '')) -> '***'
```

Challenge Time!

Challenge 2: *Factorial*

Problem: ***Factorial*** is a mathematical operation that means multiplying all the whole numbers from 1 up to a given number. For example, the factorial of 3 is 3 × 2 × 1 = 6.

Your goal: Implement `factorial(n)`, a recursive function that calculates n!.

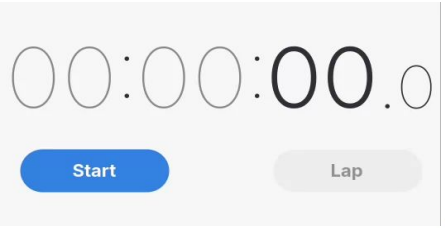
Key idea:

- **Base case:** `factorial(1) = 1`
- **Recursive case:** `factorial(n) = n * factorial(n - 1)`

Examples: `factorial(3) = 3 * 2 * 1 = 6`

`return ----------`

Challenge Time!



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- **Recursive case:** `factorial(n) = n * factorial(n - 1)`

Examples: `factorial(3) = 3 * 2 * 1 = 6`

`return ----------`

Challenge Time!

Challenge 2 Solved!

```python
def factorial(n):
    # Base case
    if n == 1:
        return 1

    # Recursive case

    return n * factorial(n - 1)



if __name__ == "__main__":
    print(f"Factorial of 3: {factorial(3)}")
```

Output:

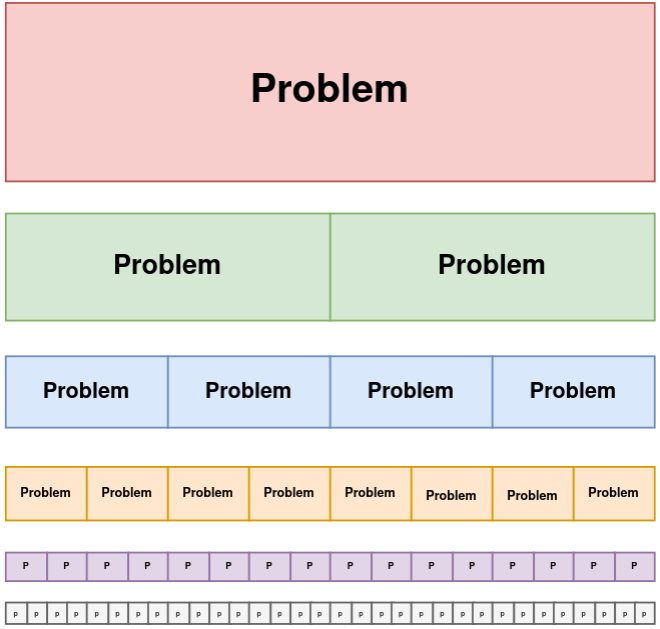
Factorial of 3: 6

Think Recursively: *Key Idea of Type 3*

- **Multiple** recursive calls
- **No partial answer** kept at each level
- Often used for search or backtracking

```
function solve(problem):
    # base case
    if problem is smallest:
        return base_answer

    # recursive case
    for smaller_problem in smaller_problems:
        solve(smaller_problem)
```

```
function solve(problem):
    # base case
    if problem is smallest:
        return base_answer

    # recursive case
    solve(smaller_problem1)
    solve(smaller_problem2)
    solve(smaller_problem3)
    ...
```



Multiple recursive calls
No partial answer

Think Recursively: *Search for a File in Folders*

- Search the **current folder**
- Search each **subfolder**
- **Stop** when the file is found or there are no more folders to search

```
function find_file(folder, target):
    # base case
    if folder is empty:
        return false
    # another base case
    if target is in folder:
        return true

    # recursive case
    for each subfolder in folder:
        if find_file(subfolder, target):
            return true

    return false
```

Think Recursively: *Escape a Maze*

- Try several **possible moves**
- Each move creates a **smaller search problem**
- **Stop** when can not move or found the exit
-

```
S = start
E = exit
# = blocked
blank = open cell
```

```
+---+---+---+---+---+---+
| S |   | # |   |   |   |
+---+---+---+---+---+---+
| # |   | # |   | # |   |
+---+---+---+---+---+---+
|   |   |   |   | # | E |
+---+---+---+---+---+---+
|   | # | # |   | # | # |
+---+---+---+---+---+---+
|   |   |   |   |   |   |
+---+---+---+---+---+---+
```

```
function can_escape(cell):
    # base case
    if cell is the exit:
        return true

    # another base case
    if cell is blocked or already visited:
        return false

    # recursive case
    mark_cell_as_visited()

    return can_escape(left) or
           can_escape(right) or
           can_escape(up) or
           can_escape(down)
```

Think Recursively: *Search a Family Tree*

- Check the current person
- Search each **child branch**
- **Stop** when the person is found or there are no more branches to search

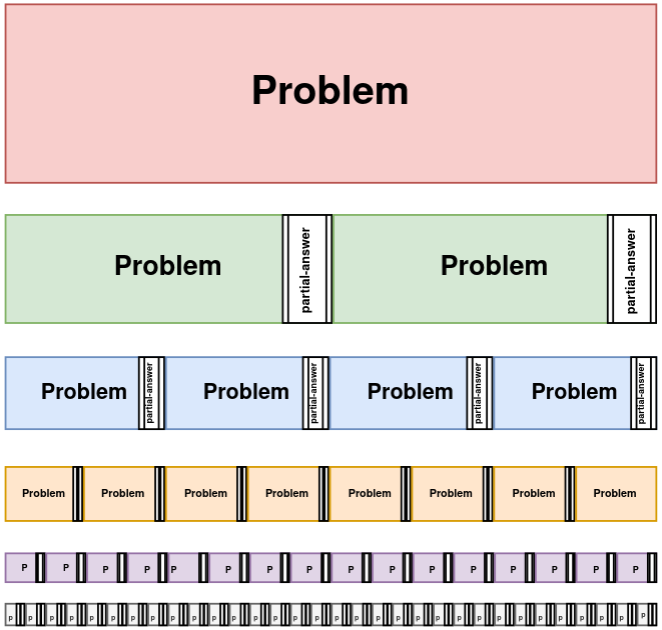
```
function find_person(person, target):
    # base case
    if person is empty:
        return false
    # another base case
    if person == target:
        return true

    # recursive case
    for each child in person.children:
        if find_person(child, target):
            return true

    return false
```

```
Family tree

          Alice
        /   |    \
     Ben   Cara   Dan
    /  \      \      \
  Eva  Finn    Gina   Hugo
             /   \
           Ivy   Jack
```

Think Recursively: *Key Idea of Type 4*

- **Multiple** recursive calls
- Each level contributes **part of the answer**
- Great for divide-and-combine problems

```
function solve(problem):
    # base case
    if problem is smallest:
        return base_answer

    # recursive case
    partial_answer = get_partial_answer(problem)
    return combine(partial_answer,
                   solve(smaller_problem1),
                   solve(smaller_problem2),
                   ...)
```



Multiple recursive calls
With partial answers

Think Recursively: *Count All Files in Folders*

- Count files in the **current folder**
- Count files in each **subfolder**
- **Stop** if folder is empty and return 0
- Add all counts together

```
function count_files(folder):
    # base case
    if folder is empty:
        return 0

    # recursive case
    total = files_in(folder)

    for each subfolder in folder:
        total = total + count_files(subfolder)

    return total
```

When to Avoid Recursion

DON’T use recursion when:

- A simple **loop is easier** to understand
- Solving the problem requires **many recursive calls**
- The recursive version is **harder to debug** and explain

When to Avoid Recursion

Avoid recursion for simple tasks such as:

- summing a list
- finding the maximum element
- scanning an array
- counting items

When to Use Recursion

Use recursion when:

- the problem **naturally breaks into smaller** subproblems
- the data has a **nested structure** (folders inside folders)
- the recursive solution is **easier** to read than the iterative one

When to Use Recursion

Use recursion, examples:

- searching folders and subfolders
- solving backtracking problems
- working with trees

```
        A
      /   \
     B     C
    / \     \
   D   E     F
      / \
     G   H
```

```python
def height(node):
    if node is None:
        return 0
    return 1 + max(height(node.left), height(node.right))
```

How to Use Recursion

Ask these three questions:

1) Is recursion better than loops here?

2) When does the function stop?

3) How does the problem get smaller?

```
def solve(problem):
    if base_case(problem):
        return answer
    smaller_problem = reduce(problem)
    return combine(solve(smaller_problem))
```

Preparing for the Final Challenge

Organize Your Mind!

Write at least 50 words about what you learned today. Feel free to include a short summary of the class for your own reference.

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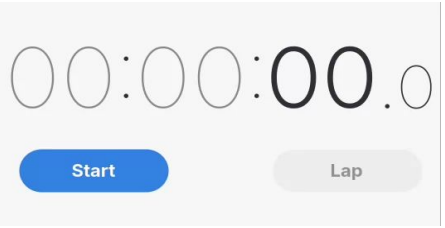
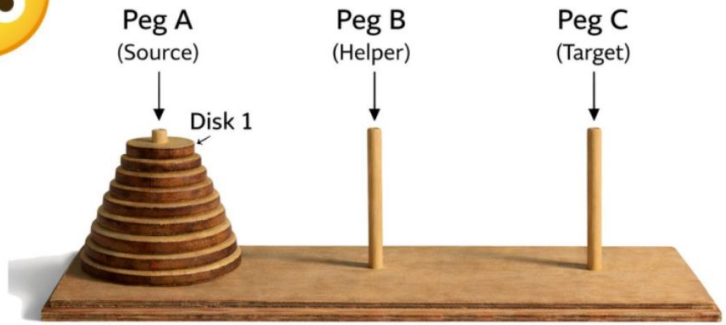
Organize Your Mind!

Write at least 50 words about what you learned today. Feel free to include a short summary of the class for your own reference.



Challenge Time!

Challenge 3: *Tower of Hanoi*



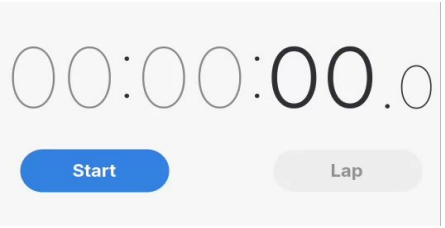
Problem: Move all disks from Peg A to Peg C

Rules:

1. Move only one disk at a time
2. You may only move the top disk from a peg
3. Never place a larger disk on top of a smaller disk

Your goal: Write a recursive function that moves **n** disks from the source peg to the target peg

Challenge Time!



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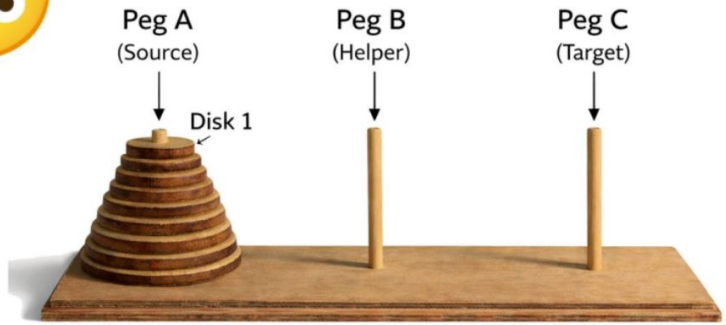

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Challenge Time!

Challenge 3 Solved!

```python
def hanoi(n, source, target, helper):
    # Base case
    if n == 1:
        print(f"Move disk 1 from {source} to {target}")
        return

    # Recursive case
    hanoi(n - 1, source, helper, target)
    print(f"Move disk {n} from {source} to {target}")
    hanoi(n - 1, helper, target, source)

if __name__ == "__main__":
    num_disks = 3
    hanoi(num_disks, 'A', 'C', 'B')
```

Challenge Time!

```
Output:
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
```

Questions?

Thank you for your attention!